successful trials (Gaussian GLM: efficiency $\sim$ representation + model_size, $\beta_{\text{topographic}} = -0.159$, $p < .001$; $\beta_{\text{textual}} = -0.116$, $p < .001$), particularly in the mid-sized models where performance is above chance and not yet saturated. Notably, the largest models approached near-optimal efficiency in this task when receiving the JSON SIR (see green continuous curve).

Furthermore, the final distance ratio (third panel of Fig. 2) also improves with scale (Gaussian GLM: final_distance_ratio $\sim$ model_size, $\beta_{\text{model\_size}} = -0.006$, $p < .001$). Smaller models (1B and 3B) fail to improve their positions in unsuccessful trials regardless of SIR type, while in mid-sized models (8B and 11B) only Cartesian representations led to improvements (Welch's t-test; Cartesian: $t = -1.96$, $p = 0.053$, marginally significant; Topographic: $t = 1.06$, $p = 0.291$; Textual: $t = 1.41$, $p = 0.162$). For the largest models (70B and 90B), all SIR types demonstrated improvements relative to the random policy (Welch's t-test; Cartesian: $t = -3.055$, $p = 0.004$; Topographic: $t = -9.223$, $p < 0.001$; Textual: $t = -7.324$, $p < 0.001$). Overall, these metrics reveal a scaling law for spatial navigation performance, and show that Cartesian representations are better tuned to convey spatial information to the models.

To gain further insight into the models' spatial reasoning, we constructed policy maps—the four right-most panels of Fig. 2—where cells represent positions relative to the goal (recentered to the middle of the grid to allow visualization), and arrows indicate the most common action chosen in each cell, colored by the relative frequency with which that action was chosen at each position in the grid. For brevity, only maps for the LLaMA-3.1-8B and LLaMA-3.2-90B models using JSON and Symbol Grid SIRs are shown. Presenting the models with a Cartesian SIR resulted in a higher proportion of correct policies across positions and model sizes, compared to other SIR types. However, larger models found the correct policy in more positions and were more consistent in their behavior. This aligns with the quantitative performance metrics, reinforcing the conclusion that Cartesian representations (and especially the JSON SIR) lead to more reliable spatial decision-making.

5 Activations Encode Spatial Features in a Mid-Sized LLM

To investigate whether specific parameters within LLMs are involved in representing spatial information, we conducted a detailed analysis of the LLaMA-3.1-8B model—the smallest variant that demonstrated above-chance performance across all SIR types (see left-most panel of Fig. 2). Our primary aim was to determine whether activations in individual layers encode the grid configuration up to a linear transformation and, if so, what are the components of that spatial model and how it depends on the SIR type.

We then trained linear regression models to predict the complete configuration of a 5×5 grid, represented as a 50-dimensional binary vector (with 25 dimensions encoding the presence or absence of the agent and 25 for the goal—details on this analysis can be found on Appendix C). The R^2 values of these linear models

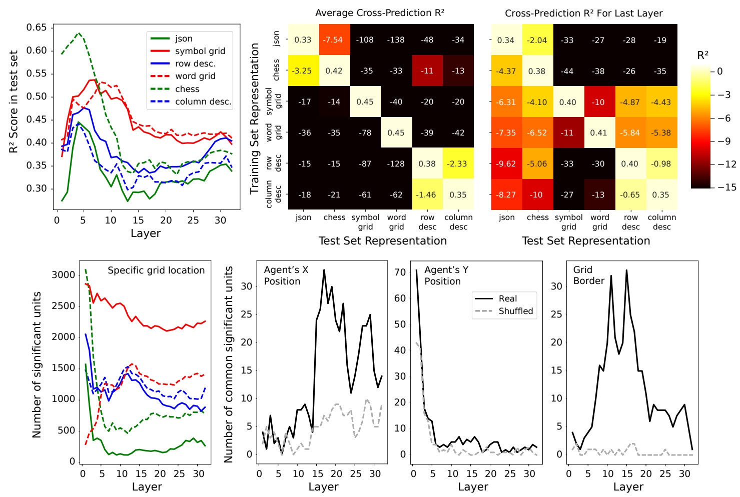

Fig. 3. Activations in LLaMA-3.1-8B predict grid configuration. Left-most top panel shows the R^2 of linear models trained on activations from individual layers to predict the full configuration of the 5×5 grid for each of the six SIR types. Second top panel shows the R^2 of linear models trained on each individual SIR type (rows) and evaluated on every individual SIR type (columns), averaged across models trained on each layer. The color scale is cut-off at -15 for visualization purposes. The third top panel shows the same cross-prediction R^2 measure but evaluated only for models trained on the last layer. The left-most bottom panel shows the number of parameters from each layer that were significantly correlated with the agent's position being in one specific cell from the 5×5 grid, for each SIR type. The last three bottom panels show the number of units in each layer that were significantly correlated with the agent's x position, y position or with border cells, for all six SIR types (black curves). Dashed gray lines shows the same calculation performed on shuffled parameter indices.

generally peaked in early middle layers and then declined in later layers, as depicted in the left-most top panel of Fig. 3. This was the case for all SIR types. However, the layer at which the R^2 peaked depended on the SIR class. For Cartesian and Textual representations, R^2 peaked around layer 5, exhibited a dip, and then showed a modest recovery in the final layers. In contrast, Topographic representations reached their maximum performance deeper in the model (around layer 10) before declining monotonically. These observations suggest that the way the model represents spatial information, and how this representation changes across layers, depends on how that information is encoded in text. Despite these differences, the test-set performance of every layer for each SIR type consistently exceeded that of a null model (Permutation Test: one per model against 10 shuffles, $p < .001$ for all cases), indicating that information about the spatial configuration of the grid is present throughout the network regardless of SIR type.

To assess the generalizability of these internal representations, we performed cross-SIR predictions by training linear models on activations derived from one SIR type and evaluating its performance on activations from the others. Averaging R^2 scores across layers revealed that representations do not generalize fully between different SIR types—performance dropped markedly when the training and testing SIRs differed (top middle panel in Fig. 3). However, this drop was considerably smaller when evaluating within the same SIR class (for example, when training on JSON and testing on Chess Notation, or between different Textual variants). As the variance of R^2 values between SIR types decreased from superficial to deep layers (left-most top panel of Fig. 3), we wondered if grid representations might also become more consistent in deeper layers. Motivated by this hypothesis, we examined cross-SIR performance using only the final layer activations (right-most top panel in Fig. 3). Indeed, we found that R^2 values were generally larger for models trained on this layer than in the average case. Furthermore, the best performance for every SIR was achieved when training and testing on the same SIR type, with the second-best performance always occurring for the other member of the same SIR class. However, only R^2 values for same-SIR testing were found to be significantly larger than those obtained by evaluating a null model (Permutation Test: one per comparison against 10 shuffles, $p < 0.05$ only for models tested on same-SIR activations). These findings suggest that the representation of grid space is mostly specific to each SIR, but there appears to be some component of that representation that is generalizable at least between SIRs of the same class. Based on this observation, we turned to look for abstract components of the model that could be representing spatial information irrespective of SIR type.

Inspired by the existence of place cells in biological systems, which activate only when an animal is in a specific position in space [44, 45], we next explored whether individual units in the model exhibit spatial selectivity with respect to the agent's position in the grid. For each SIR type, we fitted a linear model per individual parameter, using activations as regressors for predicting whether the agent occupied a specific cell of the 5×5 grid. We performed this analysis for all model parameters and for all 25 grid cells, and adjusted p-values by the number of comparisons using the Bonferroni correction (see Appendix C for more details). We found hundreds to thousands of units per layer (depending on the SIR type—see left-most bottom panel in Fig. 3) that were significantly correlated with the agent's presence in specific grid cells. However, these correlations might reflect the encoding of particular textual motifs within a SIR rather than a true, abstract spatial model. To attempt to find units that encode spatial information in a general way, irrespective of how that information is represented in the input, we looked for parameters that showed a correlation with a specific grid cell across all SIR types. Interestingly, we found no such units, suggesting that the model does not represent the position of the agent by recognizing specific spatial locations, as is the case in biological systems.

We then wondered if the agent's position in the grid might instead be encoded continuously in the x and y axes. To evaluate this, we looked for units whose